



Turning Chaos into Victory

Optimizing Random Team Composition in E-Sports Matches

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Alex Holtzapple

- Bachelor of Science – Engineering Psychology
- 12 years experience as a data analyst in several industries
- Enjoy playing games and studying both the psychological and analytics aspects of video games

Introduction

- E-sports analytics is rapidly evolving. Teams and players are increasingly leveraging data to gain a competitive advantage (Gill, 2025).
- League of Legends (LoL) - a competitive game where teams of 5 compete to destroy each others' base.
 - Typically, champions are selected by each team, and opponent selections are known before the match.
- The All Random All Mid (ARAM) game mode creates chaos by having champions assigned randomly.
 - The opponent team is not known until the match starts.
 - Players have limited options to modify their team's composition before the match.

Research Problem & Hypothesis

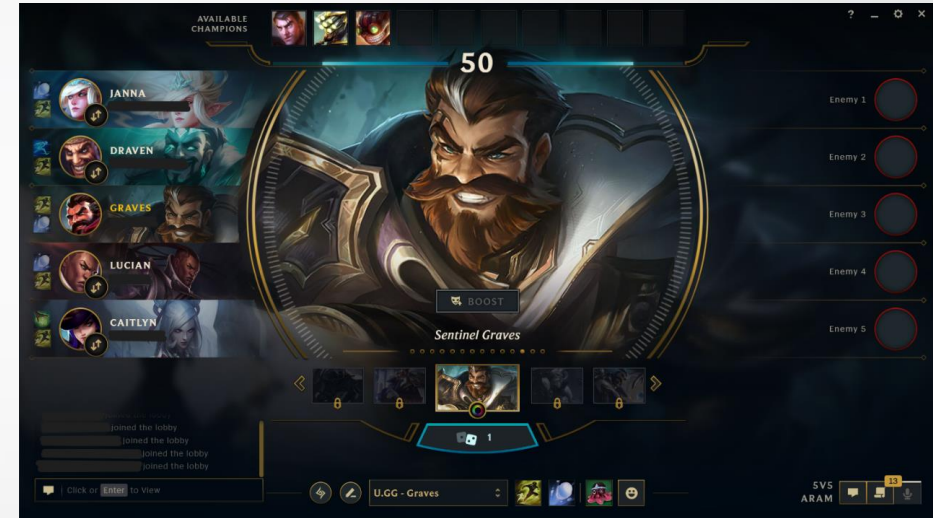
- Because of random selection, ARAM win-rates should hover around 50/50 win-loss over the long term.
 - But player intuition suggests certain compositions consistently perform better.
- This study aims to answer: *"Can a predictive model determine the winning team in an ARAM match based on team composition with an accuracy greater than random guessing (50%)?"*
 - **Null Hypothesis:** A predictive model cannot predict match outcomes better than random guessing ($\leq 50\%$ accuracy).
 - **Alternative Hypothesis:** A predictive model can predict match outcomes better than random guessing ($> 50\%$ accuracy).

ARAM game map
"Howling Abyss"



Data Collection

- Collected from the official LoL game public API. Consisted of:
 - Champion data:
 - Primary role
 - 1 to 10 Rating of: Attack, Defense, Magic, Difficulty
 - Match data:
 - Player champion selection at match start
 - Match duration
 - Team result (win / loss)
 - Filters applied:
 - Duration > 8 minutes (remove early surrenders or technical issues)
 - Maximum of 20 matches per player to avoid bias
 - Only most recent game version



ARAM champion selection screen

Data Preparation

- Data preparation steps:
 - Merged champion info with player / match data
 - One-hot encoded champion roles (Assassin, Tank, Mage, Marksman, Support, Fighter)
 - Split the single role column into new column for each role with a 1 or 0 if the champion is that role
 - Match duration was checked for outliers above a calculated limit (IQR method)
 - Matches longer than 1964.5 seconds were removed
 - Teams with rare role stackings (4 or 5 champions of the same role) were removed if they had a sample size of less than 50 teams.
 - Champion roles and ratings were aggregated for each match – team group
 - Roles were summed to produce a count of each role within a team
 - Ratings were averaged to produce an average rating for the whole team

Data Preparation

- Final dataset consisted of 11 columns and 34,375 data points:

Feature	Data Type	Description
win	Boolean (T/F), Categorical	Team result / response variable; win = True
attack_mean	Float, Continuous	Team's average champion attack rating
defense_mean	Float, Continuous	Team's average champion defense rating
magic_mean	Float, Continuous	Team's average champion magic rating
difficulty_mean	Float, Continuous	Team's average champion difficulty rating
primary_role_Assassin_sum	Int, Ordinal	Count of Assassin champions on team
primary_role_Fighter_sum	Int, Ordinal	Count of Fighter champions on team
primary_role_Mage_sum	Int, Ordinal	Count of Mage champions on team
primary_role_Marksman_sum	Int, Ordinal	Count of Marksman champions on team
primary_role_Support_sum	Int, Ordinal	Count of Support champions on team
primary_role_Tank_sum	Int, Ordinal	Count of Tank champions on team

Data Analysis – Logistic Regression Baseline

- Chosen for its specialty in predicting binary outcomes (win/loss) and easy to interpret coefficients.
- Base Model:
 - Included all predictors
 - Accuracy of 54.31%, achieving > 50% target
 - McFadden's R-Squared of only 0.009 – explained < 1% of variance in outcomes

Baseline Logistic Regression Results

Features Included:

```
Index(['attack_mean', 'defense_mean', 'magic_mean', 'difficulty_mean',  
      'primary_role_Assassin_sum', 'primary_role_Fighter_sum',  
      'primary_role_Mage_sum', 'primary_role_Marksman_sum',  
      'primary_role_Support_sum', 'primary_role_Tank_sum', 'win'],  
      dtype='object')
```

Accuracy: 0.5431

Precision: 0.5394

Recall: 0.5998

AUC-ROC: 0.5598

Confusion Matrix:

```
[[1669 1763]  
 [1378 2065]]
```

Classification	Report:				
	precision	recall	f1-score	support	
0	0.55	0.49	0.52	3432	
1	0.54	0.60	0.57	3443	
accuracy			0.54	6875	
macro avg	0.54	0.54	0.54	6875	
weighted avg	0.54	0.54	0.54	6875	

Log-Likelihood: -4723.606509299363

Null Log-Likelihood: -4765.378066345869

McFadden's R-squared: 0.008765633379963278

Durbin-Watson statistic: 1.9917223814071952

Baseline logistic regression model results

Data Analysis – Logistic Regression Final

- Implemented feature reduction method (RFE) to remove least important predictors and maintain accuracy.
- Final Model:
 - Only used difficulty rating and counts of Assassins, Mages, and Tanks
 - Accuracy of 54.55%, achieving > 50% target
 - McFadden's R-Squared of only 0.008 – explained < 1% of variance in outcomes
- **Coefficients show that Less Assassins, more Mages and Tanks lead to higher win probability.**

RFE-Reduced Logistic Regression Results

Features Included:

```
Index(['difficulty_mean', 'primary_role_Assassin_sum', 'primary_role_Mage_sum',  
      'primary_role_Tank_sum'],  
      dtype='object')
```

Accuracy: 0.5455

Precision: 0.5399

Recall: 0.6242

AUC-ROC: 0.5541

Confusion Matrix:

```
[[1601 1831]  
 [1294 2149]]
```

Classification Report:

	precision	recall	f1-score	support
0	0.55	0.47	0.51	3432
1	0.54	0.62	0.58	3443
accuracy			0.55	6875
macro avg	0.55	0.55	0.54	6875
weighted avg	0.55	0.55	0.54	6875

Log-Likelihood: -4729.461277084867

Null Log-Likelihood: -4765.378066345869

McFadden's R-squared: 0.007537028282114465

Durbin-Watson statistic: 1.96992

	Feature	Coefficient	Change in Odds
0	difficulty_mean	-0.067559	0.934672
1	primary_role_Assassin_sum	-0.153414	0.857775
2	primary_role_Mage_sum	0.099063	1.104136
3	primary_role_Tank_sum	0.047809	1.048970

Final logistic regression model results

Data Analysis – Random Forest

- Chosen for its ability to model non-linear relationships (e.g. some Mages help but too little or too many hurt)
 - Included only role counts
 - Accuracy of 54.89%, achieving > 50% target
 - AUC-ROC of 0.57, model performed slightly better than logistic regression (0.55)
 - Assassin and Mage champions were most important predictors of winning
- Example tree shows 0 Assassins, 2 Mages, 2 Tanks, 1 Marksman has highest win probability**

```
Random Forest Base Model Accuracy: 0.5489
Random Forest AUC-ROC: 0.5676
```

```
Confusion Matrix:
```

```
[[1760 1672]
 [1429 2014]]
```

```
Classification Report:
```

	precision	recall	f1-score	support
0	0.55	0.51	0.53	3432
1	0.55	0.58	0.57	3443
accuracy			0.55	6875
macro avg	0.55	0.55	0.55	6875
weighted avg	0.55	0.55	0.55	6875

```
Highest Win Probability: 0.615 (Leaf 87)
```

```
Lowest Win Probability: 0.351 (Leaf 119)
```

```
Decision Path for Highest Win Probability:
```

```
primary_role_Assassin_sum <= 0.50 → primary_role_Mage_sum > 1.50 →
primary_role_Marksman_sum > 0.50 → primary_role_Tank_sum > 0.50 →
primary_role_Mage_sum <= 2.50 → primary_role_Marksman_sum <= 1.50 →
primary_role_Tank_sum > 1.50
```

```
Decision Path for Lowest Win Probability:
```

```
primary_role_Assassin_sum > 0.50 → primary_role_Mage_sum > 0.50 →
primary_role_Marksman_sum <= 0.50 → primary_role_Fighter_sum > 0.50 →
primary_role_Assassin_sum > 1.50
```

Data Analysis – Findings

- **Both models achieved an accuracy > 50%**, providing evidence that a model using only team composition to predict ARAM wins with > 50% accuracy is possible.
- **The random forest model performed slightly better** with an accuracy of 54.89% vs. 54.55 and AUC-ROC score of 0.57 vs 0.55.
- Results show that **team composition does play a role in probability of winning**, but low accuracy suggests **there are many other factors not accounted for** in the models that significantly influence chance of victory.
- **Both models agree** on general team composition rules:
 - Assassins decrease win probability
 - Mages ,Tanks, and Marksmen improve win probability
 - Example team:
 - Galio (Tank), Garen (Tank), Jinx (Marksman), Ryze (Mage), Veigar (Mage) = **60.07% win chance**



Data Analysis – Limitations

- Features not included in analysis likely significantly contribute to win probability:
 - **Player Skill** – More skilled players could win even with bad team composition. It may be better to play a champion you know well, even if it decreases win probability according to the model.
 - **Champion Synergies** – Specific combinations could be more predictive than singular roles. Support + Marksman may be more influential as a predictor than Marksman alone.
 - **In-Game Decisions** – Items, playstyle, and communication can influence match outcomes. Team composition doesn't matter if a team doesn't work together.
 - **Opposing Team** – Teams were analyzed in isolation, not in comparison with opposing team. Knowing opponent team could improve predictability (e.g. Assassins are OK if against a team with more Supports)
 - **Single Game Version** – The game undergoes periodic updates which may change champion roles and balancing, rendering the current results obsolete.

Proposed Actions – Players & Teams

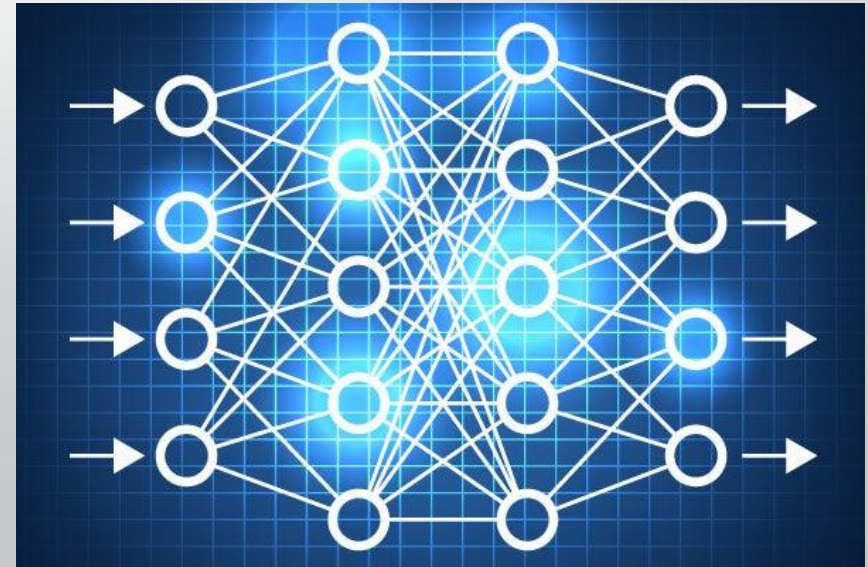
- Players and teams can benefit from the findings by taking the following actions:
 - **Ensure at least 1 Tank** – Both models show at least 1 Tank improved probability of winning.
 - **Avoid any Assassins** – Having even 1 Assassin champion showed a detrimental effect on win probability and should be avoided.
 - **Prioritize Mages** – Presence of 1 or 2 Mages consistently improved win probability. If the team is otherwise balanced, prioritize picking a Mage over other roles.
 - **Don't Be Selfish With Re-Rolls** – Re-rolls are limited and take time to replenish. They should be used only to support the above recommendations. Players shouldn't re-roll just because they didn't get the specific champion they wanted.
- Example team #2:



 - Galio (Tank), Garen (Tank), Jinx (Marksman), Ryze (Mage), Katarina (Assassin) = **48.40% win chance**
- Teams can use these models to develop an ARAM building tool that takes these recommendations into account to provide trade and re-roll guidance to increase win probability.

Proposed Actions – Future Research

- Future research should explore and incorporate the following:
 - **Bring in other variables** like player skill, item build, and other playstyle variables to try and capture the remaining variability in ARAM match outcomes
 - **Introduce complex interaction terms** (squared transforms, Support x Marksman, diversity score) to attempt to capture deeper patterns or non-linear relationships, especially for future logistic regression models.
 - **Explore other machine learning techniques.** Optimizing the models with other types of regression, hyperparameter tuning, or entirely different frameworks such as neural networks could help detect more nuanced team synergy effects.



Expected Benefits

- Incorporating the model recommendations could **increase ARAM win probability by about 5%** over random selection. Players and teams could achieve an additional 5 wins per 100 matches.
- Implementing these models at scale could **influence teams' performance at competitive ARAM tournaments**, leading to new strategies and increasing **player interest** in a mode often dismissed as purely random or “just for fun”.
- This study provides a structured approach to understanding the impact of team composition that can be **built upon in further research of League of Legends** data, or other esports titles.



Thank you!

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